

1. Outline extubation strategies for anticipated difficult extubation of an adult. (ANZCA 2017A)

Examiner Report

Pass rate 60.5%

An adequate answer needed to identify situations leading to a difficult extubation and made sensible comments on at least 2 different extubation techniques. Poorer answers focused on difficult intubation and didn't recognise the difference.

An **anticipated difficult extubation** refers to a patient who may have

- Problems with maintaining airway
 - Poor neurological status
 - Stroke, cerebral oedema, pre-op poor GCS
 - Metabolic derangements (hypoglycaemia, hyponatremia, hypothermia)
 - Aspiration risk
 - GERD, symptomatic hiatal hernia
- Problems achieving adequate ventilation and oxygenation
 - Neuromuscular disorders
 - Myasthenia Gravis
 - Suxamethonium apnoea
 - Poor respiratory status
 - Brittle asthma, severe COPD/ restrictive lung disease
 - Respiratory infection (pneumonia)
 - Poor cardiovascular status
 - High inotropic support requirement, high arrhythmic load
- Potentially difficult re-intubation
 - Known difficult airway
 - Airway deterioration intra-op
 - Major H&N surgery, airway trauma, RLN palsy, prolonged prone/ Trendelenburg position, fluid overload

Taking reference from the **UK Difficult Airway Society** extubation guidelines for 'at risk' patient, I would

Optimise patient factors

- Airway
 - Cuff leak test (>110ml drop in Tv) to exclude laryngeal oedema
 - IV Dexamethasone to reduce laryngeal oedema
 - Suction secretions from ETT/ oro-pharynx (in deep plane of anaesthesia)
 - Head position (left lateral head down if aspiration risk, head up if OSA/obese)
 - Bite-block/oral airway to avoid negative pressure oedema from spontaneous inspiration against an obstructed (bitten) ETT
- Respi
 - Pre-oxygenate patient with 100% O₂
 - Lung recruitment manoeuvre to minimise atelectasis
 - Spontaneous breathing trial with CPAP 5 cmH₂O, ensuring regular breathing pattern with adequate minute ventilation (>70ml/kg/min)
 - Consider inhaled B-agonist if concerns of bronchospasm
 - Consider Heliox/ inhaled adrenaline if concerns of upper airway obstruction
- Neuromuscular
 - Adequate neuromuscular reversal with TOF ratio > 0.9
- CNS
 - Avoidance of excessive sedative medications, long-acting opioids which may depress GCS
- CVS
 - Ensure stable haemodynamics, not on inotropes
 - Ensure judicious intra-op fluid administration (avoid overload)
 - Optimise CCF/ pulmonary oedema

- Metabolic Check and correct acidosis, hypoglycaemia, hypocalcemia
- Temperature Maintain normothermia

Optimise anaesthetic factors

- Personnel Experienced anaesthetist, and skilled help/ assistance to be present
- Environment Consider transfer to Major OT/ ICU if patient in remote location
- Equipment Equipment for re-intubation, including difficult airway trolley, to be ready.

Extubation strategies include:

1. Awake extubation with minimal stimulation and activation of airway reflexes by:
 - Providing adequate analgesia
 - Avoidance of tactile stimulation during emergence
 - Use of Lignocaine/ Remifentanyl infusion
 - Insertion of nasal airway (if the ability to maintain upper airway patency is a concern)
 - Transition to CPAP/BIPAP/HFNC post-extubation (if mechanical ventilatory support is needed)
 - Insertion of Cook Airway Exchange Catheter via ETT lumen, secured at the same ETT depth at lips (provides patent airway conduit, and facilitates intubation in a patient with known difficult intubation)
2. Alternative Strategy: Deep extubation with LMA exchange technique (Bailey Manoeuvre)
 - Involves LMA insertion after extubating patient in deep plane after ensuring adequate spontaneous respiration.
 - To facilitate smooth extubation in a patient with known anticipated difficult extubation
3. Extubation should be postponed if further optimisation of patients' factors is needed, and patient should be sent to ICU.
4. Tracheostomy, if prior discussion done with valid consent, for patient with long term risk factors for extubation failure.

Post-extubation care

- Frequent assessment and close monitoring for respiratory complications, with continuous SpO₂, RR and ETCO₂ monitoring.
- Equipment for re-intubation, including difficult airway trolley, should be present at all times in PACU.
- Adequate analgesia to facilitate lung mechanics, which needs to be balanced by minimisation of opioid related respiratory depression
- Plan for safe transfer, including supplemental O₂, to High Dependency after discharge from PACU.
- Proper documentation/ handover/communication to PACU/ HD staff